

## Light at the end of the tunnel...

X17 (2017) — English translation

In an isothermal atmosphere, the air is in equilibrium. Molar mass of air  $\mu = 29 \text{ г/моль}$ , atmospheric temperature  $T = 300 \text{ K}$ . Consider the free fall acceleration  $g$  known and constant. Also assume that the atmospheric pressure  $p_0$  at the Earth's surface is known.

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| <b>A1</b> | Derive a formula that allows you to determine the air pressure at any height $h$ relative to the level of the Earth's surface. Express your answer in terms of $p_0, \mu, g, R, T$ and $h$ . | <b>0.50</b> |
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| <b>A2</b> | Determine the height $H$ at which the pressure changes $e$ times. Express your answer using $R, T, \mu$ and $g$ . | <b>0.50</b> |
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The density distribution in the earth's crust is such that the acceleration due to gravity  $g = 9.8 \text{ м/с}^2$  is constant. The radius of the Earth is  $r_0 = 6370 \text{ км}$ . The refractive index of light in air is  $n = 1 + \Delta n$ , where  $\Delta n \ll 1$ . At the surface of the Earth  $n_0 = 1.000292$ . Over a wide pressure range  $\Delta n \sim p$ .

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| <b>A3</b> | At what depth $h$ should a tunnel be dug along the Earth's surface so that light circulates in it in a closed orbit? Consider that $h \ll r_0$ . Express your answer in terms of $R, T, \mu, g, r_0$ and $n_0$ . Calculate the resulting value $h$ . | <b>4.00</b> |
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Source: <https://pho.rs/p/308>